

BUSINESS REQUIREMENT DOCUMENT

15-Day Data Analytics Internship | Day 1

Project: E-Commerce Sales & Customer Analytics System

Role: Data Analytics Intern

Primary Tools: Excel • SQL • Python/Pandas • Power BI

1. Business Background

The project simulates the work of a Junior Data Analyst for an e-commerce company. The objective is to transform business data into reliable insights, dashboards, and management recommendations.

2. Business Problem

Management needs a clear understanding of what is driving sales and profit, who the most valuable customers are, which products and regions perform poorly, how discounts affect profitability, and what actions should be taken.

3. Project Objectives

- Analyze overall sales and profitability.
- Identify top- and bottom-performing products and categories.
- Analyze customer behavior and customer value.
- Evaluate regional sales and profit performance.
- Analyze sales trends over time.
- Evaluate the relationship between discounts, sales, and profit.
- Develop an interactive Power BI dashboard.
- Produce evidence-based business insights and recommendations.
- Present the complete project to the supervisor.

4. Stakeholders

The analysis is intended primarily for management and business decision-makers. The project will also provide useful information for sales, marketing, product, and regional business teams.

5. Key Business Questions

- What are the company's total sales, profit, orders, and customers?
- Which products and categories perform best and worst?
- Are the best-selling products also the most profitable?
- Who are the most valuable customers and how frequently do they purchase?
- Which regions generate the highest and lowest sales and profit?
- How do sales change over time?

- What is the relationship between discount, sales, and profit?
- Does larger discounting improve sales enough to justify reduced margins?

6. Required KPIs

KPI	Purpose
Total Sales	Measure overall revenue.
Total Profit	Measure absolute profitability.
Profit Margin	Measure profitability relative to sales.
Total Orders	Measure transaction volume.
Total Customers	Measure customer base.
Average Order Value	Measure average revenue per order.
Repeat Customer Rate	Measure customer retention/loyalty.
Sales Growth	Measure change in sales over time.
Top Product	Identify the strongest product by selected metric.
Regional Performance	Compare sales and profit across regions.

7. Project Scope

In scope: sales, profitability, customers, products/categories, regions, time trends, discount impact, data cleaning, SQL analysis, Python/Pandas analysis, EDA, Power BI dashboard, business insights, and recommendations.

Expected analytical flow: Understand → Clean → Analyze → Visualize → Interpret → Recommend.

8. Expected Outcomes

- A reliable cleaned dataset.
- Reproducible Excel, SQL, and Python analysis.
- Meaningful exploratory visualizations.
- An interactive Power BI dashboard suitable for management.
- Evidence-based business findings and recommendations.
- A final analytics report and presentation.